

TRINITY DATA CLEANING TOOL USER MANUAL

V 1.0

Pradyumna Shivanandaiah
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1.About Tool

The Trinity Data Cleaning Tool is a Flask-based local application that validates and cleans Excel files containing Trinity records. It automates error detection, provides suggested fixes, and allows manual review to ensure data integrity. Designed for secure, offline use, it prepares clean, standardized datasets for reporting, compliance, or system integration.

2. Purpose

The purpose of this process is to define a standardized workflow for using the Trinity Data Cleaning Tool , which ensures that Excel files containing Trinity records are validated, cleaned, and corrected in a consistent and efficient manner. The tool automates error detection based on predefined rules and allows users to manually review and fix discrepancies before generating a finalized, clean dataset. This helps maintain data integrity, reduce manual effort, and support downstream operations such as reporting, analysis, or integration with other systems.

3. Scope

This process applies to all individuals or teams who are required to validate and clean Excel-based datasets containing Trinity records, particularly when these datasets are used for internal reporting, compliance, or system migration purposes. It outlines how to run the Flask-based local application, upload encrypted or non-encrypted Excel files, perform validation checks, and apply fixes either automatically or manually.

This documentation covers:

- Starting the Flask application locally.
- Accessing the web-based validation portal.
- Uploading and validating Excel files.
- Interpreting and resolving errors.
- Downloading the final processed file.

This process does not cover:

- Backend development or maintenance of the Flask application.
- Advanced troubleshooting of Python or Flask-related issues (refer to Github Repo)

4. Start Trigger

The process is initiated when any of the following conditions occur:

- A new Excel file containing Trinity records needs to be imported into a system that requires structured and validated data.
- An existing file has been flagged for inconsistencies or format errors during previous processing.
- A scheduled data refresh or batch update requires preparation of clean input files.
- A user receives an Excel file from an external source and must verify its integrity before further use.

In essence, the process begins whenever there is a need to ensure that an Excel file conforms to the expected structure and quality standards defined by the Trinity Data Cleaning Tool .

5. End Result / Outcome

At the conclusion of this process, the user will have:

- A validated Excel file that meets all structural and formatting requirements.
- A clear log of detected errors, along with options to ignore , manually correct , or automatically fix each issue.
- A cleaned and corrected output file ready for use in downstream applications, databases, or reports.
- Optionally, a record of changes made during the validation process, enabling traceability and audit readiness.

If no errors are found, the tool will confirm that the file is already valid and provide a downloadable copy without modifications.

6. Step-by-Step Process

Step 1: Open the Flask Application Folder

1. Navigate to the folder where the Flask application is stored.
2. Right-click on the folder and select Open in Terminal (or equivalent based on your operating system).

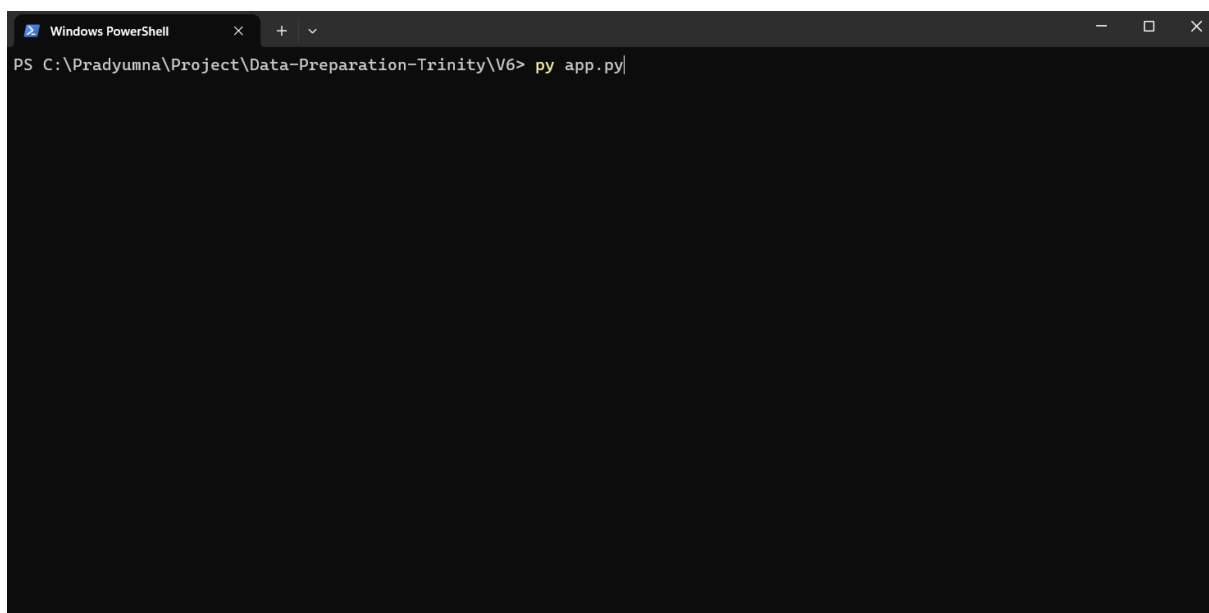
Example Path:

C:\Pradyumna\Project\Data-Preparation-Trinity\V6

Step 2: Start the Flask Application

1. In the terminal, enter the following command to start the Flask application:

```
py app.py
```



- Note: Ensure that Python is installed and accessible in your terminal.
- The terminal will display output similar to the following:

```
* Serving Flask app 'app'
* Debug mode: off
INFO:werkzeug:WARNING: This is a development server. Do not use it in a production deployment.
Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8000
* Running on http://172.16.0.2:8000
INFO:werkzeug:Press CTRL+C to quit
```

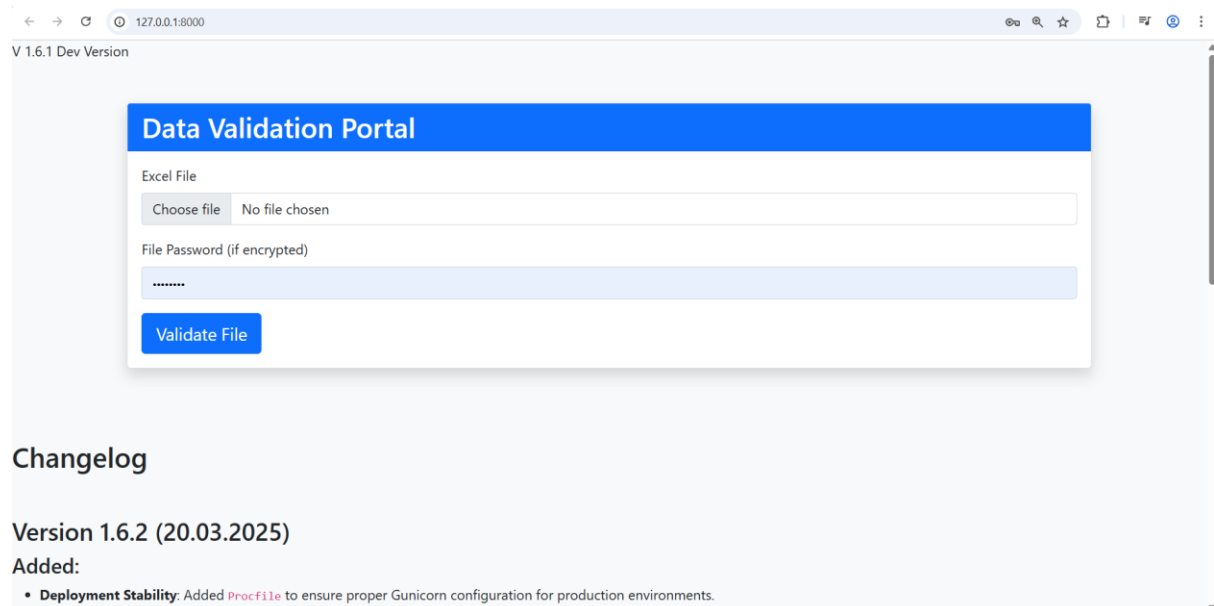
```
Windows PowerShell
PS C:\Pradyumna\Project\Data-Preparation-Trinity\V6> py app.py
* Serving Flask app 'app'
* Debug mode: off
INFO:werkzeug:WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8000
* Running on http://172.16.0.2:8000
INFO:werkzeug:Press CTRL+C to quit
INFO:werkzeug:127.0.0.1 - - [27/May/2025 16:11:17] "GET / HTTP/1.1" 200 -
```

Step 3: Access the Validation Portal

1. Open a web browser.
2. Enter one of the URLs displayed in the terminal:

```
http://127.0.0.1:8000
http://172.16.0.2:8000
```

3. The Data Validation Portal will open.



Step 4: Upload the Excel File

1. On the Data Validation Portal , locate the section for uploading the Excel file.
2. Click the Choose file button and select the Excel file containing Trinity records.
3. If the Excel file is encrypted, enter the password in the File Password (if encrypted) field.

Step 5: Validate the File

1. After selecting the file and entering the password (if applicable), click the Validate File button.
2. The tool will process the file and perform all predefined validations.

Step 6: Review Validation Results

1. Once the validation is complete, the Validation Results page will display:
 - Type: Indicates the type of error (e.g., Structure, Validation).
 - Row: The row number where the error occurred.
 - Field: The column or field name where the error was found.
 - Error Reason: A description of the issue.
 - Current Value: The value currently in the file.
 - Corrected Value: Suggested correction (if applicable).
 - Action: Options to Ignore or fix the error.

Validation Results for Week_10_Import_File-Master_File.xls

Type	Row	Field	Error Reason	Current Value	Corrected Value	Action
Structure	-	-	Missing columns: Welcome Call, Channel. Suggestion: Add these columns to your file.	-	No input required for structural errors.	
Validation	281	IBAN	Invalid IBAN format: IE07IPBS99064352495842aa	IE07IPBS99064352495842aa	IE07IPBS99064352495842aa	<button>Ignore</button>
Validation	281	Eircode	Invalid Eircode format: PP1209DO (expected format: A11 AA11)	PP1209DO	PP1209DO	<button>Ignore</button>
Validation	281	District	Invalid District: Malway (not found in database)	Malway	Malway	<button>Ignore</button>

Apply Fixes & Download Back to Upload

Validation Rules in Trinity Data Cleaning Tool with Common Fix:

Validation Type	Field	Error Message Sample	Example Invalid Input	Suggested Fix
Structure	Any	Missing columns: Welcome Call	Column missing	Add the column
Structure	Any	Unexpected columns found: Middle Name	Extra column	Remove the column
Validation	IBAN	Invalid IBAN format: DE89370400440547798673	German IBAN	Use IE29AIBK93603312345678 or ignore
Validation	BIC	Invalid BIC format: ABCD1234	Short BIC	Use AIBKIE2D
Validation	Date	Invalid date format: 12/05/1990	DD/MM/YYYY	Use 1990-05-12
Validation	Date	Missing date value	Blank cell	Enter valid date
Validation	Opt-In	Invalid opt-in value: Yes	'Yes'	Use '1' or '0'
Validation	Eircode	Invalid Eircode format: Dublin 4	Text address	Use D02 X009
Validation	District	Invalid District: Atlantis	Unknown name	Use valid CRM district

Step 7: Apply Fixes or Ignore Errors

1. For each error:

- If the suggested Corrected Value is accurate, leave it as is.
- If the error should be ignored, click the Ignore button

2. After reviewing all errors:

- Click the Apply Fixes & Download button to apply all corrections and download the cleaned file.

- Alternatively, click Back to Upload to re-upload the file or make changes.

Validation Results for Week_10_Import_File-Master_File.xls

Type	Row	Field	Error Reason	Current Value	Corrected Value	Action
Structure	-	-	Missing columns: Welcome Call, Channel. Suggestion: Add these columns to your file.	-	No input required for structural errors.	
Validation	281	IBAN	Invalid IBAN format: IE07IPBS99064352495842aa	IE07IPBS99064352495842aa	IE07IPBS99064352495842aa	Ignore
Validation	281	Eircode	Invalid Eircode format: PP1209DO (expected format: A11 AA11)	PP1209DO	PP1209DO	Ignore
Validation	281	District	Invalid District: Malway (not found in database)	Malway	Malway	Ignore

Step 8: Download the Cleaned File

- After applying fixes, the tool will generate a cleaned version of the Excel file.
- Download the file for further use.

Frequently Asked Questions (FAQ)

Q: What if the Excel file is not encrypted?

A: Simply skip the File Password (if encrypted) field and proceed with uploading the file.

Q: Can I ignore all errors?

A: Yes, but ignoring errors may result in incomplete or incorrect data. It's recommended to review each error before deciding to ignore it.

Q: Why am I getting a warning about the development server?

A: The warning indicates that the Flask application is running in development mode. This is normal.

Q: How do I stop the Flask application?

A: Press **CTRL+C** in the terminal to stop the application.